AKHILA ELIZABETH JOSE

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Aerospace engineer with expertise in electric propulsion, spacecraft structures, and numerical modeling. Proficient in CAD (CATIA, SolidWorks), CFD, FEA/FEM (ANSYS), and MATLAB for propulsion simulations. Graduate research centered on analyzing how thermal dissociation affects performance of iodine-fed Hall thrusters, with additional work on oscillation damping via RLC filter integration. Eager to apply simulation-driven expertise and multidisciplinary skills in propulsion, modeling, and structural design to advance innovative solutions for space missions and sustainable exploration.

EDUCATION

- **Master of Science in Space Engineering** – *University of Pisa (UNIPI), Italy*| 2024-2025
- **Bachelor of Technology in Aerospace Engineering-** *Amity University (U.P.), India*| 2014-2018
- **Schooling -** *The Aditya Birla Public School (M.P.), India* 2000-2014

PROJECT EXPERIENCE

- **Thesis: Analysing the role of thermal dissociation on the performance of Iodine fed Hall Thruster –** *University of Pisa, Italy*| 2024-2025
  - Analyzed the suitability of **iodine as an advanced propellant for the SPT-100 Hall thruster, emphasizing its potential to outperform xenon in space applications.**
  - Developed an **RLC circuit model** to **dampen high-frequency oscillations** caused by **plasma instabilities** and integrated it with the 1-D fluid model
  - Conducting **MATLAB-based simulations** for **current variations, oscillation behavior, and plasma discharge stability**.
  - Investigated the **role of thermal dissociation in iodine-fed Hall thrusters** and its influence on performance parameters, **validating numerical simulations against expected experimental behavior** to identify **performance-optimization strategies.**
- **Vega Stage 3 Structural Analysis**-*Spacecraft Structures Project* **-** *University of Pisa, Italy* | Oct 2023- Jul 2024
  - Conducted preliminary **structural analysis of Zefiro 9A**, the third stage of the Vega launch vehicle.
  - Designed the **solid propellant fuel tank** using **CATIA & SOLIDWORKS**
  - Performed **structural, modal, and vibration analysis in ANSYS**
  - Assessed stress distribution, deformation, and shock response spectra.
- **Io exploration Mission** | *Space Systems Project* **-** *UNIPI, Italy* | Sep 2022 - May 2023
  - Defined mission objectives and operational constraints for a deep space mission targeting Io, Jupiter's volcanic moon.
  - Conducted **mission analysis**, subsystem trade-offs and **budget assessments**.
  - Designed two impactor probes and modelled plume-surface interactions using **Smoothed Particle Hydrodynamics.**
  - Coordinated **risk assessment and mitigation strategies**, identifying potential mission risks and formulating solutions for subsystem integration and performance optimization.
  - Developed the mission architecture, subsystem configuration, and integration strategy, with supporting documentation and final technical review.
- **Self-Healing Composite** | *Major Project (Amity University)* **|** Dec 2017 - May 2018
  - Led a team of four members to research and develop a Self-Healing Composite.
  - Designed and **3D modeled Self-Healing Composite** and performed structural analysis using Ansys.
  - Optimized Self-Healing Composite material capable to use as an aerospace-grade material.
  - Demonstrated the working principle of the optimized Self-healing composites at a macroscopic level.

INTERNSHIP EXPERIENCE

- **Brahmastra Aerospace**| *Virtual CubeSat Internship* Nov 2020 - Dec 2020
  - Led a team of five members to work on "**Structural Design and Propulsion System of CubeSats.**"
  - Proposed various methods of green propulsion and reviewed the advancement of innovation in the field.

- **Hindustan Aeronautics Limited**| *Bangalore* | June 2017 - July 2017
  - Analyzed manufacturing processes and assembly of aerospace-grade components for aircraft structures and propulsion systems, gaining hands-on exposure to industry-standard techniques

- **Cetpa Infotech Internship**| *Noida* | May 2016 - June 2016
  - Received training in Design, assembly and drafting using Cativa V5.
  - Designed **combustion chamber and turbine** with various airfoil configurations to optimize efficiency.

CONFERENCES AND SEMINARS

- **Modern Research in Aerospace Engineering** | *Amity University* | *Noida, India* Sept 2017
- **Nasa Spaceward India** | *Amity University* | *Noida, India* Aug 2016

SKILLS

- **Technical**
  - **CAD & Structural Design:** CATIA V5, SolidWorks, ANSYS Geometry & Meshing
  - **Numerical Modelling:** CFD (ANSYS Fluent), FEA/FEM (ANSYS), SPH (Beginner)
  - **Programming:** MATLAB,Simulink, Python, C, C++, LaTeX
  - **Software & Tools:** Microsoft Office, Adobe Photoshop, Illustrator, Krita
- **Languages**
  - Hindi (Native), Malayalam (Native), English (Fluent), Italian (Beginner)
- **Soft Skills-** Problem Solving, critical thinking, time management, creativity, teamwork

HOBBIES

Art and Crafts	Music- Playing Ukulele
Photography, videography and editing	Indian Classical Dance- Bharatnatyam, Kathak